



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO B R N And Sons Fuel

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

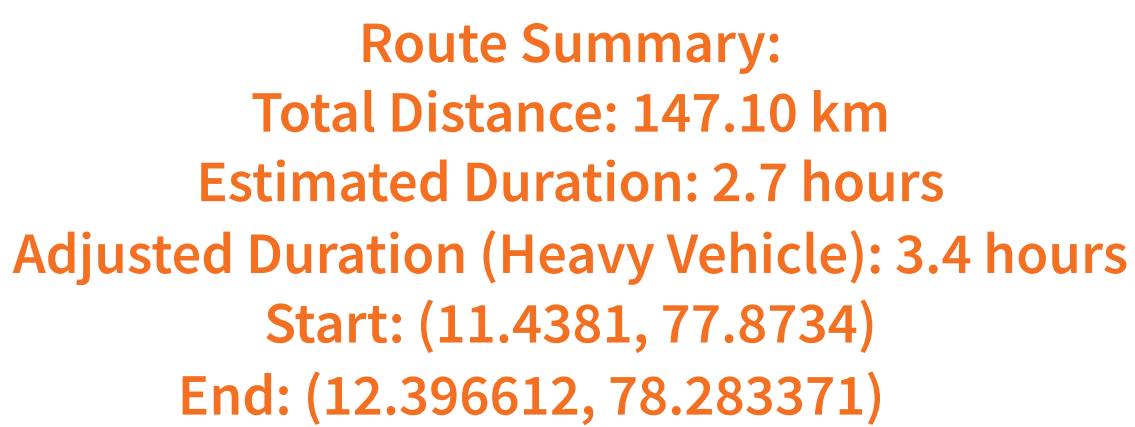
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



1. Overview of the Route Map

The journey from Sangagiri to Nagojanahalli spans approximately 147.10 kilometers, primarily following national and state highways that connect these regions. Major points on this route include sections of NH44 and SH20, which are integral for navigating between the two locations.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Patterns:** Tamil Nadu experiences a tropical climate with hot summers, mild winters, and significant monsoon rainfall from October to December.
- **Hazards:** Monsoon season may cause flooding or waterlogging, potentially impeding travel and affecting road conditions. Drivers should also be prepared for reduced visibility during heavy rains and heat during peak summer months.

3. Traffic Patterns and Congestion-Prone Areas

- **Peak Hours:** Traffic congestion typically peaks in morning (8-10 AM) and evening (5-7 PM) due to local commuting.
- **Congestion Hotspots:** Urban centers and entry points of major towns, such as Erode and Salem, often experience higher traffic density.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** The main highways (NH44 and SH20) are generally well-maintained, but care should be taken on smaller connecting roads which may have potholes or uneven surfaces.
- **Infrastructure:** Vulnerable bridges or narrow passages might require attention. Regular updates from local transport authorities can provide current statuses.

5. Suggestions for Alternative Routes for Emergencies

- **Primary Alternative:** An alternative can be taken via NH544, bypassing tramplng urban centers.
- **Secondary Options:** Local roads connecting neighboring towns might serve as detours, although they might be longer and slower.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu follows stringent guidelines for the transport of hazardous materials, including:

- Compliance with the Motor Vehicles Act and central regulations.
- Restrictions on movement through residential or sensitive areas.
- Requirement of specific permits and labeling for hazardous goods.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

Historical data indicates occasional incidents involving heavy vehicles, with common causes being driver error, mechanical failure, and adverse weather conditions. Notably, junctions and sharp turns have been flagged as higher-risk zones.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** Proximity to natural reserves or historically significant spots where restrictions may apply.
- **Environmental Concerns:** High emissions and risk of spillage in ecologically sensitive zones necessitate heightened vigilance.

9. Analysis of Communication Coverage and Potential Dead Zones

Most of the route maintains robust cellular coverage with occasional dead zones in rural stretches. Consider using satellite communication systems or pre-downloaded maps for these sections.

10. Estimated Emergency Response Times for Different Route Segments

- **Urban Areas:** Rapid response, generally within 15-30 minutes due to better infrastructure.
- **Rural Areas:** Emergency response might take 45 minutes to an hour, depending on the distance from major towns.

11. Overall Summary of Risk Assessment

The route from Sangagiri to Nagojanahalli is feasible for the transport of hazardous materials with proper preparation. Key risks include weather conditions, traffic congestion, and specific road challenges. Comprehensive planning and adherence to regulations are critical to mitigate these issues.

Key Recommendations:

- Stay updated on weather forecasts, particularly during monsoon months.
- Plan travel outside peak hours to avoid congestion.
- Monitor real-time traffic updates and road conditions via reliable navigation apps.
- Ensure compliance with all transportation and environmental regulations.
- Prepare for emergency situations by knowing alternative routes and keeping emergency contacts readily available.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.46400, 77.87323	30 KM/Hr
5	Turn	Medium	11.46765, 77.87037	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	High	11.49149, 77.89601	15 KM/Hr
8	Turn	High	11.49224, 77.89601	15 KM/Hr
9	Turn	Medium	11.53501, 77.95707	30 KM/Hr
10	Turn	Medium	11.53601, 77.95856	30 KM/Hr
11	Turn	Medium	11.62403, 78.10893	30 KM/Hr
12	Turn	High	11.64180, 78.11808	15 KM/Hr
13	Turn	High	11.64325, 78.11993	15 KM/Hr
14	Turn	Medium	11.69519, 78.10775	30 KM/Hr
15	Turn	Medium	11.73958, 78.05105	30 KM/Hr
16	Turn	Medium	11.75835, 78.05016	30 KM/Hr
17	Turn	Medium	11.86847, 78.08367	30 KM/Hr
18	Turn	Medium	11.87942, 78.08197	30 KM/Hr
19	Turn	Medium	11.92585, 78.06507	30 KM/Hr
20	Turn	Medium	11.93704, 78.05402	30 KM/Hr
21	Turn	High	11.95818, 78.04978	15 KM/Hr
22	Turn	Medium	11.95988, 78.04948	30 KM/Hr
23	Turn	Medium	11.96304, 78.05136	30 KM/Hr
24	Turn	High	11.96399, 78.05135	15 KM/Hr
25	Turn	High	11.96580, 78.04923	15 KM/Hr
26	Turn	Medium	11.96675, 78.04939	30 KM/Hr
27	Turn	Medium	11.96791, 78.05070	30 KM/Hr
28	Turn	Medium	12.19881, 78.18157	30 KM/Hr
29	Turn	Medium	12.22459, 78.19136	30 KM/Hr
30	Turn	Medium	12.25060, 78.19977	30 KM/Hr
31	Turn	Medium	12.25329, 78.20039	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
32	Turn	Medium	12.29110, 78.19909	30 KM/Hr
33	Blind Spot	Blind Spot	12.40692, 78.21762	10 KM/Hr
34	Turn	High	12.40626, 78.22057	15 KM/Hr
35	Turn	Medium	12.40692, 78.22160	30 KM/Hr
36	Turn	High	12.40667, 78.22271	15 KM/Hr
37	Turn	Medium	12.40763, 78.22338	30 KM/Hr
38	Turn	Medium	12.41678, 78.22485	30 KM/Hr
39	Blind Spot	Blind Spot	12.41727, 78.22529	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
11	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
15	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
19	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
20	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
21	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
22	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
24	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
25	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
26	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
27	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
28	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
31	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
32	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
33	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
34	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
35	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
37	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
13	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
14	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
23	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
29	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
30	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
36	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
38	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62403, 78.10893



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64180, 78.11808



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64325, 78.11993



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69519, 78.10775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73958, 78.05105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75835, 78.05016



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86847, 78.08367



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92585, 78.06507



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93704, 78.05402



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95988, 78.04948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19881, 78.18157



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22459, 78.19136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25060, 78.19977



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25329, 78.20039



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.29110, 78.19909



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.40692, 78.21762



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.40626, 78.22057



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40692, 78.22160



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.40667, 78.22271



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40763, 78.22338



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.41678, 78.22485

Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.41727, 78.22529

